

Technology Stack

Date	01 OCTOBER 2026
Team ID	6ab7f7bfd083b0e2ee72b13b
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	2 Marks

Define Your Technology Stack:

The technology stack for LegalEase identifies the programming language, frontend and backend frameworks, AI service, databases, document-formatting libraries, deployment infrastructure and supporting tools used to build the solution.

The choices below are based on the supplied Legal Ease project documentation and preserve the technologies documented there.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend/ ClientSide	Streamlit, HTML/Streamlit styling	Streamlit provides the web interface through which users enter document requirements, view generated legal content, preview and edit the document, and access download options. HTML/Streamlit styling supports presentation and user-interface formatting.
2	Backend / ServerSide	Python, FastAPI, Pydantic, Uvicorn	Python handles the application logic and document-generation workflow. FastAPI provides the API layer, routing and request handling. Pydantic validates structured document requests, while Uvicorn runs the FastAPI application.

3	Database / Data Storage	SQLite; Firebase Cloud Firestore	SQLite is documented for lightweight structured application data and local development. Firebase Cloud Firestore is documented as a cloud NoSQL option for user data, document metadata, saved templates and generation history.
4	Cloud / Hosting / Deployment	Uvicorn + Streamlit locally; Render, Railway, Deta, Streamlit Community Cloud, Fly.io or VPS for future deployment.	The supplied documentation specifies local deployment with Uvicorn for the FastAPI backend and Streamlit for the frontend. It lists Render, Railway, Deta, Streamlit Community Cloud, Fly.io and VPS deployment as possible future cloud options.
5	Version Control & CI/CD	Git + GitHub	Git is used to track changes in the LegalEase source code and maintain different versions of the project. GitHub provides a remote repository for storing the code, collaborating with team members, managing branches, and maintaining project history.
6	Third-Party APIs / Other Tools	Google Generative AI / Gemini API; python-docx; FPDF	The Google Generative AI / Gemini API is used to generate legal-document content from structured prompts. python-docx and FPDF convert the generated legal text into downloadable DOCX and PDF documents, while TXT output is also supported.